

Requirements Traceability Matrix (RTM) - Final

Req ID	Requirement Description	Test Case ID	Test Scenario & Expected Result	Test Status
1. Donor Module (Mobile & Web)				
FR-D1	Donor shall be able to register with name, age, phone, address, blood type, gender, and location.	TC-D01	Verify successful registration when a donor inputs all valid required data.	● Passed
FR-D2	Donor shall login using username and password.	TC-D02	Verify successful login with correct credentials and rejection of invalid ones.	● Passed
FR-D3	Donor shall update his/her profile (including availability and last donation date).	TC-D03	Verify donor can successfully modify profile details and save changes.	● Passed
FR-D4	Donor shall receive push notification when a nearby hospital needs his/her blood type.	TC-D04	Verify FCM triggers a push notification to the donor's device upon a matching request.	● Passed
FR-D5	Donor shall view details of the blood request (hospital name, location, contact).	TC-D05	Verify notification opens a detailed view displaying accurate hospital request data.	● Passed
FR-D6	Donor shall view information page about donation rules and eligibility.	TC-D06	Verify the donation rules/eligibility page loads correctly and is readable.	● Passed
2. Hospital Module (Web)				
FR-H1	Hospital shall register with name, phone, address, and location.	TC-H01	Verify hospital account is created successfully pending admin approval.	● Passed
FR-H2	Hospital shall login.	TC-H02	Verify authorized hospital personnel can log in successfully.	● Passed
FR-H3	Hospital shall create a blood request (blood type needed, quantity, urgency).	TC-H03	Verify system accepts and broadcasts a new blood request with proper parameters.	● Passed
FR-H4	Hospital shall view list of nearby compatible donors with distance.	TC-H04	Verify UI displays a sorted list of compatible donors based on geographic proximity.	● Passed

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FR-H5	Hospital shall contact donors directly via the displayed phone number.	TC-H05	Verify donor contact details are accurately displayed to the hospital user.	Passed
3. Admin Module (Web)				
FR-A1	Admin shall login with special credentials.	TC-A01	Verify admin dashboard access is strictly granted to admin credentials.	Passed
FR-A2	Admin shall approve/reject hospital registrations.	TC-A02	Verify admin can successfully change hospital status to 'Approved' or 'Rejected'.	Passed
FR-A3	Admin shall delete donors or hospitals.	TC-A03	Verify admin possesses full CRUD capabilities for users.	Passed
FR-A4	Admin shall monitor all blood requests and their status.	TC-A04	Verify admin dashboard displays a real-time, accurate list of all active/past requests.	Passed
FR-A5	Admin shall send confirmation message to donor after donation.	TC-A05	Verify system successfully dispatches confirmation messages post-donation.	Passed
FR-A6	Admin shall send reminder to donor after 3 months from last donation.	TC-A06	Verify system auto-triggers or allows admin to trigger 3-month reminders.	Passed
FR-A7	Admin shall generate reports (number of donors, requests, blood types availability).	TC-A07	Verify admin can export accurate statistical reports based on system data.	Passed
4. Core System Features				
SYS-01	Smart Compatibility Engine: match donors and recipients beyond identical types.	TC-S01	Verify system suggests O- donors for A- requests based on the compatibility matrix.	Passed
SYS-02	Automated Deferral Logic: enforce 3-month waiting period and medical rejection.	TC-S02	Verify system blocks donation requests for users who donated within the last 90 days.	Passed
SYS-03	Automatic matching of blood type and location (using Haversine formula).	TC-S03	Verify the calculated distance between hospital and donor coordinates is accurate.	Passed

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SYS-04	Security: Password not visible during registration (masked input).	TC-S04	Verify that the password field input is masked (e.g., asterisks) during registration.	Passed
5. Non-Functional Requirements				
NFR-P1	System shall respond to requests within 5 seconds.	TC-N01	Verify API response times remain under 5 seconds during load testing.	Passed
NFR-P2	Notification shall be delivered within 30 seconds of request creation.	TC-N02	Verify FCM latency between hospital request submission and donor receipt is <30s.	Passed
NFR-S1	All passwords shall be stored using strong hashing (bcrypt/PHP password_hash).	TC-N03	Verify database inspection shows passwords stored purely as bcrypt hashes.	Passed
NFR-S2	Role-based access control (Donor / Hospital / Admin).	TC-N04	Verify access control restricts endpoints appropriately based on user role.	Passed
NFR-S3	Protection against SQL injection and XSS.	TC-N05	Verify input fields sanitize malicious scripts and SQL queries during testing.	Passed
NFR-S4	Medical Data Privacy: All clinical screening results and health history must be encrypted.	TC-N06	Verify HIPAA-compliant encryption standards are applied to medical history tables.	Passed
NFR-R1	System shall be available 24/7.	TC-N07	Verify server uptime and redundancy configurations.	Passed
NFR-R2	Database backup shall be performed daily.	TC-N08	Verify automated daily cron jobs execute and store SQL database dumps securely.	Passed
NFR-U1	Interface shall be simple/intuitive for all age groups.	TC-N09	Verify UI/UX passes user acceptance testing (UAT).	Passed